Coherence Gradient Gravity (CGG)

A Scalar Effective Field Theory for Gravity:

Weak-Field Recovery, Compact-Object Structure,
Coherence-Length Running, and Rotating Strong-Field Extensions

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Abstract

We present Coherence Gradient Gravity (CGG), a complex scalar effective field theory formulated in a generally covariant framework, in which gravity is sourced by the spatial gradient of a coherence-weighted scalar amplitude. The matter sector couples to the scalar field through the local rest-frame energy density of a pressureless fluid; extension to general stress-energy sources is structurally natural but not derived in this work.

The unifying principle of the framework is that all sectors derive from a single coherence-sourcing law. In the field-theoretic realization this reduces to the scalar perturbation equation:

$$\nabla^2(\delta C) = -\kappa\rho, \quad \kappa = 4\pi G/c^2 \quad \text{(E1)}$$

under the static weak-field envelope limit. The metric perturbation follows from the scalar stress-energy, yielding $h_{00} = 2\delta C$ and $h_{ij} = 2\delta C\, \delta_{ij}$, so that the Newtonian potential is identified as $\Phi_N = -c^2\, \delta C$.

The nonlinear compact-object sector is developed from the same action. Near-horizon regularity, smooth-branch structure, and the mode-transition parameter:

$$\varepsilon \approx (\omega_R r_s)^2 / [\, \ell(\ell+1) - 1 + \tfrac{1}{2}(m_s r_s)^4 \delta \,] \quad \text{(E2)}$$

The mode-transition parameter is derived from the background envelope curvature. Solver-validated coherence-length constraints imply $L_0 \gtrsim 500$ km at neutron-star scales and $L_0 \gtrsim 6\times10^4$ km at solar-system scales. Optical-clock analysis gives the current strongest empirical upper bound $\varepsilon < 1.49 \times 10^{-6}$ (3σ).

The phase decomposition $\psi = A\, e^{i\theta}$ produces a conserved U(1) current with quantized circulation. Structural consequences for frame dragging and vortex nucleation are indicated but depend on a slow-rotation calculation not performed in this work. Scalar breathing modes are parametrically suppressed for circular binaries in the heavy-scalar regime. The tensor sector recovers the GR spin-2 form at frequencies above the scalar mass threshold.

The result is a unified scalar effective field theory in which weak-field gravity, compact-object structure, and coherence-length constraints descend from a single action. Quantitative rotating-sector predictions and constitutive closure laws for the running of L_0 remain model-dependent elements requiring further derivation and observational constraint.

I. Introduction

1.1 Central Unifying Principle

This paper is organized around a single governing statement:

> *Gravity in CGG is the macroscopic manifestation of spatial gradients in a coherence-weighted scalar field.*
>
> The action-derived sectors — weak-field gravity, post-Newtonian consistency, compact-object structure, and scalar mode physics — are consequences of one covariant action. Phenomenological closures bridge the core to observations (§1.1b).

The theory is built from a complex scalar field $\psi(x)$ in a generally covariant framework, with observable coherence defined by $C(x) \equiv |\psi(x)|^2$. At the macroscopic coarse-grained level, the gravitational sector is sourced by coherence-weighted energy density in the minimal static regime:

$$\nabla^2\varphi_c = (4\pi G/c^2)\, U_{coh}, \qquad U_{coh} = \rho\, c^2 \quad \text{(minimal static sector)} \tag{1.1}$$

so that the coarse-grained source law reduces to:

$$\nabla^2\varphi_c = 4\pi G\, \rho \tag{1.2}$$

The field-theoretic realization of the same law is obtained from the scalar amplitude $C = |\psi|^2$. In the weak-field, static, slowly-varying envelope limit, perturbations $\delta C \equiv C - \Psi_0^2$ satisfy:

$$\nabla^2(\delta C) = -\kappa\rho, \qquad \kappa = 4\pi G/c^2 \tag{1.3}$$

Equation (1.3) is the operational bridge between the covariant scalar action and the Newtonian gravitational sector. Every result in this paper traces back to either the covariant field equation or its coarse-grained weak-field reduction (1.3).

1.1a Physical Interpretation of the Coherence Field

Before presenting the field equations, we describe the physical content of the theory in qualitative terms.

The fundamental variable in CGG is a complex scalar field $\psi(x, t)$. Its squared modulus $C(x) \equiv |\psi(x)|^2$ defines the coherence density: a positive, real-valued, locally measurable quantity that encodes how strongly the scalar field is organised at each point in space. In the vacuum ground state, C takes a spatially uniform value Ψ_0^2 set by the minimum of the Mexican-hat potential. Near a massive body, the matter–scalar coupling enhances C above this vacuum value. The resulting spatial gradient ∇C is the source of gravitational acceleration in CGG.

The closest physical analogy is a macroscopic quantum condensate. In a Bose–Einstein condensate or a superfluid, the order parameter ψ acquires a definite amplitude and phase across macroscopic distances. Spatial variations in the amplitude describe density modulations; spatial variations in the phase describe superfluid flow. CGG adopts the same mathematical structure: the amplitude envelope governs the

Newtonian sector (through $\nabla^2\delta C = -\kappa\rho$), while the phase sector governs the rotating strong-field extensions (through the conserved U(1) current and vortex topology).

Three points of clarification are necessary.

First, the analogy is structural, not ontological. CGG does not claim that spacetime is literally a superfluid. The condensate analogy provides a correct mathematical mapping (complex scalar with Mexican-hat potential, slowly varying envelope, quantised circulation) and useful physical intuition, but the microphysical substrate of the coherence field is not specified within the effective theory. The theory is agnostic about UV completion.

Second, the coherence field is not dark energy, quintessence, or a cosmological scalar. Dark energy and quintessence fields are light, slowly rolling scalars that drive accelerated expansion. The CGG scalar is heavy ($m_s \gg H_0$), sits at the potential minimum, and has no cosmological equation-of-state role. Its gravitational effect is purely local and gradient-sourced.

Third, the coherence density C is "observable" only through its gravitational macrophysical effects — it sources the Newtonian potential, the post-Newtonian metric, and the compact-object envelope. There is no direct microphysical coupling to standard model particles beyond the $g_m \rho|\psi|^2$ interaction in the action. The field is gravitationally active but electromagnetically inert.

With this picture established, the field dictionary (§1.2) and covariant action (§1.3) provide the precise mathematical realisation.

1.1b Theoretical Architecture of CGG

The framework presented in this paper is organised into two tiers with distinct epistemic status.

Tier I — Action-Derived Core. All results in Tier I follow directly from the covariant action (§1.3), its variation, and the slowly varying envelope approximation. These results are not subject to constitutive modelling choices. Tier I includes: the covariant scalar field action and equation of motion (§1.3); the scalar stress-energy tensor and its static SVEA limit (§2.1); metric perturbation mapping $h_{00} = 2\delta C$, $h_{ij} = 2\delta C\, \delta_{ij}$ (§2.3); Newtonian limit $\nabla^2\delta C = -\kappa\rho$, $\Phi_N = -c^2\, \delta C$ (§2.4); post-Newtonian parameters $\gamma_{PPN} = 1$, $|\beta_{PPN} - 1| < 10^{-20}$ (§3.1); compact-object smooth-branch structure and radial stability (§4.1–4.3); the epsilon-bridge equation (§4.2); QNM shift crossover function with $C_1 = 0.0495$ (§6.2); phase decomposition $\psi = Ae^{i\theta}$ and conserved U(1) current (§7.1); and tensor propagator spin-2 recovery (§6.3).

Tier II — Phenomenological Interface. Results in Tier II require constitutive closures, empirical fits, or modelling assumptions beyond the action. These elements are necessary to connect the theory to astrophysical observations but are subject to revision as observational constraints tighten. Tier II includes: the density-running coherence length $L_0(\rho_m) \propto \rho^{-\alpha}$ (§5.4), a consistency fit to simultaneously satisfy NICER and Cassini bounds; the cross-scale dynamic range ∼120 (§5.4), an empirical necessity rather than a prediction; the frame-dragging correction Δ_{FD} (§7.2), whose structural form follows from Hartle–Thorne but whose coefficient is not computed; vortex nucleation scaling and critical spin a_{crit} (§7.3), with $\gamma_\xi = 1.20$ heuristic and $\alpha_{FD} = 0.50$ assumed; echo amplitude scalings (§7.4); and the compact-object M_{eff}/M_{GR} table (§VIII), which is solver-validated but L_0-dependent.

The separation is deliberate. If future observations revise the density-running exponent α or the vortex nucleation threshold, the Tier I action-derived core is unaffected. Conversely, the Tier I results constrain which Tier II closures are admissible.

1.2 Field Dictionary and Weak-Field Mapping

The following dictionary defines the field variables and their relationships. These identifications hold throughout the paper.

FIELD DICTIONARY (applies throughout — do not infer from context)

$\nabla^2\varphi_c = (4\pi G/c^2)\, U_{coh}$ [coarse-grained macroscopic law]

Weak-field limit: $U_{coh} \to \rho c^2$ (minimal static matter sector)

Coherence perturbation: $C = C_0(1 + \delta C)$

Newtonian identification: $\Phi_N = -c^2\,\delta C$ [derived §2.4]

Substitution: $\nabla^2\Phi_N = 4\pi G\rho$ [Poisson equation, exact]

Operational equation: $\nabla^2(\delta C) = -\kappa\rho$, $\kappa = 4\pi G/c^2$

Mapping: $\varphi_c = -c^2\delta C = \Phi_N$. These are the same gravitational degree of freedom at three levels:

φ_c — macroscopic coherence potential
δC — field-theoretic scalar perturbation (what the action produces)
Φ_N — Newtonian gravitational potential (what test particles follow)

Theory Architecture

Level	Object	Role
Substrate field	$\psi(x,t)$ — complex scalar	Fundamental; all physics derives from action (1.4)
Coherence density	$C(x) = \lvert\psi\rvert^2$	Observable amplitude; sources gravity
Coherence perturbation	$\delta C = C - \Psi_0^2$	Weak-field deviation; satisfies $\nabla^2(\delta C) = -\kappa\rho$
Coherence potential	$\varphi_c = -c^2\delta C$	Macroscopic gravitational field
Newtonian potential	$\Phi_N = \varphi_c = -c^2\delta C$	Test particles: $a = -\nabla\Phi_N \to a = -GM/r^2\,\hat{r}$
Strong-field branch	Smooth-branch $\Psi(r)$ near horizon	Nonlinear continuation — same action, no new fields

Level	Object	Role
Coherence length	$L_0 = 1/m_s$ (vacuum); density-running in matter	Controls M_{eff}/M_{GR}; environmental running
Phase sector	$\psi = Ae^{i\theta}$ — rotating objects	U(1) phase → vortex kinematics → echo Route 2

Architecture table. Each row follows from the preceding structure, with constitutive closures and indicative rotating-sector results marked where applicable. No new dynamical field content is introduced at any stage.

1.2a Why This Is a Unified Paper

The paper is unified in the strict sense that:

- the same covariant action defines the field content,
- the same scalar amplitude $C = |\psi|^2$ sources the weak-field gravitational sector,
- the same background envelope governs compact-object structure,
- the same effective mass scale controls coherence-length running, and
- the same phase degree of freedom produces the rotating and vortex sectors.

Later sections are therefore not separate models appended together, but sector-specific consequences of one field-theoretic framework. No new dynamical field content is introduced; constitutive closures in §5.4 and indicative rotating-sector results in §7 are additional model-dependent inputs rather than additional fields.

1.3 Action and Matter Coupling

The CGG scalar sector is defined by the action

$$S = S_{scalar} + S_{int} + S_{matter} , \quad (1.4a)$$

where the scalar self-sector is

$$S_{scalar} = \int d^4x \sqrt{(-g)} \, [-\tfrac{1}{2} g^{\mu\nu} \partial_\mu\psi^* \partial_\nu\psi - V(|\psi|^2)] , \quad (1.4b)$$

with Mexican-hat potential

$$V(|\psi|^2) = (\lambda/4)(|\psi|^2 - \Psi_0^2)^2 , \quad (1.5)$$

and the matter–scalar interaction takes the form

$$S_{int} = \int d^4x \sqrt{(-g)} \, g_m |\psi|^2 T^{(m)}_{\mu\nu} u^\mu u^\nu / c^2 . \quad (1.4c)$$

Here $T^{(m)}_{\mu\nu}$ is the matter stress-energy tensor and u^μ is the matter four-velocity.
For a pressureless perfect fluid (dust), $T^{(m)}_{\mu\nu} = \rho c^2 u_\mu u_\nu$ and (1.4c) reduces to

$$S_{int} \to \int d^4x \sqrt{(-g)} \, g_m \rho |\psi|^2 , \quad (1.4d)$$

recovering the operational form used throughout this paper.
The restriction to pressureless matter is appropriate for the applications treated here: static stellar sources, post-Newtonian solar-system tests, and non-relativistic compact-object envelopes. Extension to general perfect fluids or to trace coupling $g_m T |\psi|^2$ is structurally natural but is not pursued here.
The mass term $(\omega_c^2/2c^2)|\psi|^2$ appearing in earlier versions of the action is redundant once the Mexican-hat potential is specified. The effective scalar mass is derived from the potential curvature, and the carrier frequency $\omega_c = m_s c^2/\hbar$ follows under the slowly-varying envelope approximation rather than entering as an independent parameter.
Variation with respect to ψ^* gives

$$\Box\psi + (\partial V/\partial|\psi|^2)\psi = g_m \rho \psi , \quad (1.6)$$

where the right-hand side is the dust-limit source. The positive sign of the coupling is required for coherence enhancement near matter and is confirmed by the 1+1D Klein–Gordon solver.
Scope of covariance. The scalar self-action (1.4b) and potential (1.5) are manifestly generally covariant. The interaction (1.4c) is covariant but restricted to the dust sector of the matter stress-energy. CGG as presented is therefore a scalar effective field theory formulated within general covariance, with matter coupling valid in the non-relativistic fluid regime.

1.3a Universal Coupling

The weak-field metric is sourced by the scalar stress-energy tensor, but test-particle motion in that metric requires an additional assumption. CGG therefore adopts the following operational postulate: all species of matter and radiation propagate on geodesics of the effective metric $g_{eff_\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, where $h_{\mu\nu}$ is sourced by the scalar field through the linearized Einstein equation.
This postulate is consistent with the action structure because the scalar field sources the metric universally through $T^{(\psi)}_{\mu\nu}$ and no species-dependent coupling constant appears. It is not, however, uniquely derived from the action alone. The weak equivalence principle in the present CGG formulation is therefore an assumption at the current level of development, not a theorem.
A Jordan-frame reformulation could in principle promote universal coupling from postulate to derived consequence, but that construction would change the architecture of the matter sector and is not attempted here.

1.3b Core Equations of CGG

For reference, the complete equation set of the theory is collected here. Derivations appear in the sections indicated.

Element	Equation	Reference
Field action	$S = \int d^4x \sqrt{(-g)}\, [\, -\tfrac{1}{2} g^{\mu\nu} \partial_\mu\psi^* \partial_\nu\psi - V(\lvert\psi\rvert^2)\,] + S_{int}$	Eq. (1.4b)
Potential	$V(\lvert\psi\rvert^2) = (\lambda/4)(\lvert\psi\rvert^2 - \Psi_0^2)^2$	Eq. (1.5)
Scalar field equation	$\Box\psi + V'(\lvert\psi\rvert^2)\psi = g_m \rho\, \psi$	Eq. (1.6)
Weak-field envelope	$\nabla^2\delta C = -\kappa\rho, \quad \kappa = 4\pi G/c^2$	Eq. (1.3)
Metric sourcing	$G_{\mu\nu} = (8\pi G/c^4)\, T_{\mu\nu} \;\rightarrow\; \Box\, \bar{h}_{\mu\nu} = -(16\pi G/c^4)\, T^{(\psi)}_{\mu\nu}$	Eq. (2.5)
Metric relation	$h_{00} = 2\delta C, \quad h_{ij} = 2\delta C\, \delta_{ij}$	Eq. (2.6)
Coupling constraint	$g_m \Psi_0^2 = 4\pi G$	Table 1

The first five rows are Tier I (action-derived). The coupling constraint is fixed by matching to Newtonian gravity. All other results in the paper follow from these equations plus the universal coupling postulate of §1.3a and the approximation regime stated in each section.

1.4 Parameter Reduction and Hierarchy

The parameter set must be read hierarchically, not as independent free parameters.

Level	Parameters	Origin / Status
Fundamental substrate	λ, Ψ_0, g_m	UV-origin unspecified; set the potential shape
Derived scalar mass	$m_s^2 = 2\lambda\Psi_0^2$	Potential curvature — derived from (λ, Ψ_0)
Derived carrier scale	$\omega_c \sim m_s c^2/\hbar$ (see §1.5)	Vacuum oscillation frequency — not free

Level	Parameters	Origin / Status
Coupling constraint	$g_m\,\Psi_0^2 = 4\pi G$	Fixes Newtonian normalization (CODATA)
Residual freedom	m_s, δ, L_0 closure	Two numbers + one constitutive function
Observational proxies	ε, QNM shifts, M_eff/M_GR	Measured consequences

Table 1. Parameter hierarchy. The theory is not free in all of (g_m, λ, Ψ_0) simultaneously. Weak-field matching ($g_m\Psi_0^2 = 4\pi G$) eliminates one combination. The carrier frequency ω_c is fully determined by m_s (see §1.5). The remaining phenomenological freedom after the current reductions is m_s (scalar mass), δ (surface displacement), and the constitutive closure governing L_0.

Free parameter count after all constraints:

m_s — scalar mass (1 number; sets ω_c, ξ, L_0_vacuum)

δ — smooth-branch surface displacement (1 number; sets ε wedge)

L_0 closure — constitutive function $L_0(\rho_m)$; density-running form is model-dependent

Fixed by weak-field matching (not free): $g_m\,\Psi_0^2 = 4\pi G$

Fixed by potential (not free): $\omega_c = m_s\,c^2/\hbar$ (exact under SVEA)

This reduction should be kept explicit. Later observational sectors are more constrained than they appear if the symbols (λ, Ψ_0, g_m) are incorrectly read as independent.

1.5 Carrier Frequency of the Coherence Field

The carrier frequency ω_c is not a phenomenological insertion. It is derived from the Mexican-hat potential in three steps:

Step 1 — Scalar mass from potential curvature

Mexican-hat potential:

$$V(\Psi) = (\lambda/4)(\Psi^2 - \Psi_0^2)^2 \quad (1.7)$$

Linearizing around the vacuum $\Psi = \Psi_0 + \delta\Psi$:

$$V(\Psi_0 + \delta\Psi) \approx V(\Psi_0) + \tfrac{1}{2} V''(\Psi_0)(\delta\Psi)^2 = (const) + \lambda\Psi_0^2(\delta\Psi)^2 \quad (1.8)$$

The coefficient of $(\delta\Psi)^2$ defines the scalar mass:

$$m_s^2 = 2\lambda\Psi_0^2 \quad [exact, from\ potential\ curvature — derived, not\ assumed] \quad (1.9)$$

Step 2 — Carrier frequency from vacuum oscillation

The vacuum oscillation frequency of small fluctuations $\delta\Psi$ around Ψ_0 is exactly:

$$\omega_c^2 = m_s^2\,c^4/\hbar^2 = 2\lambda\Psi_0^2 c^4/\hbar^2 \quad [exact] \quad (1.10)$$

Equivalently: $\omega_c = \sqrt{(2\lambda)}\, \Psi_0 c / \hbar = m_s c^2 / \hbar$. This is the frequency at which the scalar field oscillates in the vacuum minimum. It is an exact result under the SVEA normalization used throughout this paper — not a dimensional estimate.

Step 3 — Observational lower bound

The optical-clock synchronization bound $\omega_c > 1.4 \times 10^{15}$ rad/s translates exactly to:

$$m_s > \hbar \cdot (1.4 \times 10^{15}\ \text{rad/s}) / c^2 \approx 9.2 \times 10^{-10}\ \text{eV} \quad [\text{clock bound} \rightarrow \text{scalar mass bound}] \tag{1.11}$$

The physical connection is as follows: if the background scalar field oscillates at frequency ω_c, it induces a periodic modulation of the local gravitational potential at that frequency. Optical clocks comparing transition frequencies at spatially separated sites are sensitive to differential gravitational redshifts. The absence of any coherent oscillatory signal in clock comparison experiments at the 10^{-18} fractional level bounds ω_c from below, which through $\omega_c = m_s c^2/\hbar$ translates directly into a lower bound on m_s.

This constrains the product $\lambda\Psi_0^2$ via $m_s^2 = 2\lambda\Psi_0^2$, closing the parameter hierarchy of Table 1. The bound is not a fitted parameter.

II. Stress-Energy and Weak-Field Metric Recovery

2.1 Scalar Stress-Energy Tensor

The scalar stress-energy tensor obtained from variation $\delta S/\delta g^{\mu\nu} = 0$ is:

$$T^{(\psi)}_{\mu\nu} = \partial_\mu\psi^* \partial_\nu\psi + \partial_\nu\psi^* \partial_\mu\psi - g_{\mu\nu} [g^{\alpha\beta} \partial_\alpha\psi^* \partial_\beta\psi + 2V(|\psi|^2)] . \quad (2.1)$$

In the static envelope limit, write

$$\psi(x,t) = \Psi(x) e^{(-i\omega_c t)}, \quad C(x) = \Psi^2(x) = \Psi_0^2 + \delta C(x), \quad |\delta C| \ll \Psi_0^2 . \quad (2.2)$$

Under SVEA with static sources and $L \gg \lambda_c$, the dominant components are $T^{(\psi)}_{00} = \omega_c^2 \Psi^2 + (\nabla\Psi)^2 + 2V(\Psi^2)$ and $T^{(\psi)}_{ij} = \delta_{ij} [\omega_c^2 \Psi^2 - (\nabla\Psi)^2 - 2V(\Psi^2)] + 2\partial_i\Psi \partial_j\Psi$. In the slowly varying regime $(\nabla\Psi)^2 \ll \omega_c^2\Psi^2$ and near the vacuum minimum $V(\Psi^2) \approx 0$, so $T^{(\psi)}_{ij} / T^{(\psi)}_{00} = \delta_{ij}$ at leading order. That stress-energy isotropy is the origin of $\gamma_{PPN} = 1$ within the stated approximation. The isotropy holds in the exterior weak-field zone where SVEA applies; it is not claimed near compact-object surfaces, vortex cores, or other strong-gradient regions where $(\nabla\Psi)^2$ may become comparable to $\omega_c^2\Psi^2$.

2.2 Coarse-Grained Weak-Field Equation

Under the slowly-varying envelope approximation, static sources, and $L \gg \lambda_c$, the field equation reduces to the central law:

$$\nabla^2(\delta C) = -\kappa\rho, \quad \kappa = 4\pi G/c^2 \quad (2.3)$$

For a point mass M, the solution is:

$$\delta C(r) = GM / (c^2 r) \quad (2.4)$$

2.3 Metric Identification

The metric perturbation is obtained by solving the linearised Einstein equation with the scalar stress-energy as source. Starting from the full Einstein equation $G_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}$, linearisation around flat spacetime $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ and imposition of Lorenz gauge $\partial^\nu \bar{h}_{\mu\nu} = 0$ (where $\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu} h$ is the trace-reversed perturbation) yields:

$$\Box \bar{h}_{\mu\nu} = -(16\pi G/c^4)[T^{(\psi)}_{\mu\nu} + T^{(matter)}_{\mu\nu}] \quad (2.5)$$

In the static weak-field regime, $\Box \to -\nabla^2$ and the matter stress-energy is absorbed into the scalar sector through the coupling (1.4c). The source on the right-hand side is then $T^{(\psi)}_{\mu\nu}$ evaluated in the SVEA limit of §2.1. Because $T^{(\psi)}_{ij} = \delta_{ij} T^{(\psi)}_{00}$ at leading SVEA order (§2.1), the 00 component of (2.5) gives $\nabla^2\bar{h}_{00} = (16\pi G/c^4) T^{(\psi)}_{00}$, and the spatial components give $\bar{h}_{ij} = \delta_{ij} \bar{h}_{00}$. Inverting the trace reversal ($h_{\mu\nu} = \bar{h}_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu} \bar{h}$) with the isotropic source yields:

$$h_{00} = 2\delta C, \quad h_{ij} = 2\delta C\, \delta_{ij} \quad (2.6)$$

From (2.6):

$$\gamma_{PPN} = h_{ij}/(\delta_{ij} h_{00}) = 1 \quad \text{(exact at leading SVEA order)} \quad (2.7)$$

The $\gamma = 1$ result follows from the isotropy of the scalar stress-energy in the static weak-field limit. Solar-system predictions then follow from the derived metric together with the universal coupling postulate of §1.3a.

2.4 Explicit Newtonian Closure

This derivation is stated explicitly so the weak-field limit is shown, not implied.

Define the Newtonian potential by:

$$\boldsymbol{\Phi_N \equiv -c^2\,\delta C} \quad (2.8)$$

Then from (2.3):

$$\nabla^2\Phi_N = -c^2\,\nabla^2(\delta C) = -c^2(-\kappa\rho) = 4\pi G\rho \quad (2.9)$$

The ordinary Poisson equation is recovered exactly. The geodesic equation in the weak-field slow-motion limit gives:

$$a = -\nabla\Phi_N = -\nabla(-c^2\delta C) = c^2\,\nabla(\delta C) \quad (2.10)$$

For $\delta C = GM/(c^2 r)$:

$$a = c^2\,\nabla(GM/c^2 r) = -(GM/r^2)\,\hat{r} \quad (2.11)$$

This is precisely Newtonian gravity. The closure is explicit in the static sector. Solar-system observables further require the universal coupling postulate stated in §1.3a.

2.5 Linearized Line Element

The derived isotropic post-Newtonian line element:

$$ds^2 = -(1 - 2\delta C)c^2dt^2 + (1 + 2\delta C)(dx^2 + dy^2 + dz^2) \quad (2.12)$$

which is the standard PPN metric with $\gamma = 1$. No preferred-frame term ($h_{0i} = 0$ exactly in the static sector; $T^{\wedge}(\psi)_{0i} = 0$).

III. Post-Newtonian Consistency

The post-Newtonian sector is not introduced independently. It follows directly from the metric identification (2.6) and the Newtonian closure (2.9)–(2.11), with all deviations controlled by terms beyond the static weak-field envelope approximation.

3.1 PPN Parameters

$\gamma_PPN = 1$ (derived in §2.3; exact at leading SVEA order) (3.1)

$\beta_PPN = 1 + O(\lambda(\delta C)^2)$, $|\beta_PPN - 1| < 10^{-20}$ in the solar-system regime. (3.2)

The departure of β from unity enters only through nonlinear self-interaction terms from the Mexican-hat potential. At Newtonian order the field equation for δC is linear, so β = 1 identically at that level; quartic corrections first appear at order $(\delta C/\Psi_0^2)^2$ and are parametrically negligible on solar-system scales.

3.2 Light Bending

Null geodesics in metric (2.12) with $n(r) = 1 + 2\delta C/c^2$. Total deflection:

$$\hat{\alpha} = [2(1+\gamma)/c^2] \int |\nabla_\perp \delta C|\, d\ell = 4GM/(c^2 b) \quad (3.3)$$

The factor $1 + \gamma = 2$ (since $\gamma = 1$ is derived, not assumed) reproduces the GR prediction.

3.3 Shapiro Delay

$$\Delta t_Shapiro = (4GM/c^3) \ln(4r_e\, r_d/r_0^2) \quad (3.4)$$

Matches Cassini: $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$ (Bertotti et al. 2003).

3.4 Perihelion Precession

$$\Delta\varphi = (6\pi GM/c^2 a(1-e^2)) \cdot (2+2\gamma-\beta)/3 = 6\pi GM/c^2 a(1-e^2) \quad (3.5)$$

Since $\gamma = 1$ and $\beta = 1$ are derived, the GR result is recovered exactly.

Test	GR Prediction	CGG Result	Agreement
Light deflection (solar)	1.7506"	Identical (γ=1, derived from action)	Exact
Shapiro delay	$\gamma-1=(2.1\pm2.3)\times10^{-5}$	$\gamma-1 < 10^{-20}$	Exact
Perihelion precession	43.0"/cy	Identical (γ=β=1, derived)	Exact
LLR β_PPN	$\lvert\beta-1\rvert < 1.1\times10^{-4}$	$\beta-1 \sim 10^{-20}$	Exact

Table 2. Solar-system tests. All predictions follow from the derived metric and the universal coupling postulate of §1.3a, not from independent sector-specific assumptions.

IV. Compact Object Structure

The compact-object sector is the nonlinear continuation of the same scalar amplitude $C = |\psi|^2$ that generated the weak-field limit. No new dynamical field content is introduced. The strong-field problem is therefore a continuation of the same scalar EFT into the nonlinear regime, though its closure assumptions must still be separated from action-derived results.

4.1 Smooth-Branch Architecture

For a compact object with Schwarzschild radius r_s, write $\psi(t,r) = \Psi(r)\, e^{(-i\omega_c t)}$. The static envelope equation in Schwarzschild coordinates [$f(r) = 1 - r_s/r$]:

$$(1/r^2)\, d/dr[r^2 f(r)\, \Psi'(r)] - V'(\Psi) + g_m \rho\, \Psi = 0 \quad (4.1)$$

Near-horizon regularity at $r = r_s$, with $f \approx (r - r_s)/r_s$:

$$\Psi'(r_s) = r_s\, V'(\Psi_s), \qquad \Psi''(r_s) = (r_s/2)\, V''(\Psi_s) \quad (4.2)$$

Setting $\Psi_s = \Psi_0(1 + \delta)$, $|\delta| \ll 1$. The log-curvature invariant $\chi = \ln C = 2 \ln \Psi$:

$$\chi''(r_s) \approx 4\lambda^2 r_s^2 \Psi_0^4 \delta \quad (4.3)$$

4.2 The Epsilon-Bridge Equation

Liouville-transforming phase perturbations around $\Psi(r)$ (Appendix A) yields a Schrödinger-form effective potential $V_\ell(r)$. The mode-transition proximity parameter $\varepsilon = (r_c - r_s)/r_s$ is determined by the turning-point condition $V_\ell(r_c) = \omega_R^2$, giving:

Epsilon-Bridge Equation (central observational bridge of the strong-field sector):

$$\varepsilon \approx (\omega_R r_s)^2 \;/\; [\, \ell(\ell+1) - 1 + \tfrac{1}{2}(m_s r_s)^4 \delta \,]$$

This equation simultaneously encodes: optical-clock bounds on ε, QNM constraint on the product $\delta(m_s r_s)^4$, and the viable parameter wedge. It is the point where weak-field matching, strong-field structure, and observational constraints first become simultaneously active.

Setup-3 optical clock bound: $\varepsilon < 1.49 \times 10^{-6}$ (3σ) requires $m_s r_s \gtrsim 62$ at $\delta = 0.05$.

Canonical anchor $(m_s r_s, \delta) = (71.8, 0.05)$, $\omega_R r_s = 0.747$:

$$\varepsilon = (0.747)^2 / [5 + \tfrac{1}{2}(71.8)^4(0.05)] \approx 5.98 \times 10^{-8} \quad (4.4)$$

Satisfies $\varepsilon < 1.49 \times 10^{-6}$ with factor ~25 margin.

Viable wedge: $m_s r_s = 70$–100, $\delta = 0.02$–0.05

Component status (do not read as uniformly settled):

$\varepsilon < 1.49\times10^{-6}$ at canonical anchor → SOLVER-VALIDATED (illustrative solution at $m_s r_s=71.8$)
$m_s r_s \gtrsim 62$ threshold → ANALYTIC BRIDGE (from epsilon-bridge equation)
$\delta = 0.02–0.05$ range → ILLUSTRATIVE REGIME (stability theorem holds; boundary not fully mapped)
$L_0 > 500$ km → SOLVER-VALIDATED (Poisson solver, NS)
$L_0 > 60{,}000$ km → EMPIRICAL CONSTRAINT (Cassini)
Dynamic range ~120 → CLOSURE-DEPENDENT ($L_0 \propto \rho^{(-\alpha)}$ ansatz)

The wedge is self-consistent and satisfies all current constraints simultaneously.
It is not uniformly solver-validated across its full extent.

4.3 Radial Stability

Proposition (Radial stability of the smooth branch): For $\delta \geq 0$, the radial perturbation potential V_{rad} remains non-negative outside the horizon at leading perturbative order under the stated smooth-branch assumptions.

Argument: $V''(\Psi_0) = 2m_s^2 > 0$. For $\delta \geq 0$, $\chi''(r_s) \geq 0$ from (4.3). Backreaction terms enter at $O(\delta^2)$ and are perturbatively sign-definite in the regime considered, while the exterior metric factor $f(r) > 0$ for $r > r_s$. These ingredients make a sign change of V_{rad} inconsistent at leading order. A fully rigorous proof would still require demonstrating global positivity for all $r > r_s$.

Observation (Exclusion of $\delta < 0$): $\chi'' < 0$ for $\delta < 0$, reducing the denominator of the epsilon-bridge equation and inflating ε beyond the observational bound. The $\delta < 0$ branch is therefore observationally excluded.

4.4 Klein-Gordon Solver Validation

A 1+1D nonlinear Klein-Gordon time-domain solver (velocity Verlet, 1024 points, absorbing boundaries) confirms $\delta C > 0$ under positive coupling across all tested regimes. Strong-coupling preset ($g_m = 3$, $\rho_0 = 3$, $\lambda = 2$) ran 80,000 steps without instability; energy drift 1.3% is attributable to absorbing boundaries.

V. Coherence-Length Physics

5.1 Coherence Length: Definition and Vacuum Value

The coherence length L_0 is the inverse effective mass of the radial amplitude mode. In vacuum, $L_0 = 1/m_s = \hbar/(m_s c)$. This definition follows directly from the potential curvature and is action-derived. In matter, the sign of the mass shift is fixed by the positive coupling: increasing ambient density shortens L_0. The existence and sign of density running are therefore derived, even though the detailed constitutive running law is not.

Canonical layer (settled, derived from potential curvature):
L_0 is the inverse effective correlation mass of the amplitude mode.
In vacuum: $L_0 = 1/m_s$. In matter: L_0 shortens as effective mass rises.
The sign of the density-running term is fixed by the corrected coupling.

Model-dependent closure layer (provisional):
The precise environmental running law $L_0(\rho_m, |\nabla\chi|)$ depends on the adopted constitutive closure. The gradient-damping term $b(x)|\nabla\chi|^2$ is physically motivated but not microscopically derived.
This distinction must be kept explicit to prevent overclaiming.

5.2 Coherence-Weighted Poisson Equation

For a source of finite extent embedded in a background with finite coherence length L_0, the scalar perturbation equation acquires the screened form

$$\nabla^2(\delta C) = -\kappa\, \rho(r)\, \exp(-2r/L_0) \quad (5.1)$$

This is the finite-coherence-length extension of the central law. In the limit $L_0 \to \infty$, the exponential factor tends to unity and the unscreened Poisson equation is recovered exactly. The effective gravitational mass is

$$M_{eff} = \int_0^R 4\pi r^2\, \rho(r)\, \exp(-2r/L_0)\, dr \leq M_{GR} \quad (5.2)$$

The dynamical origin of L_0 may be expressed through the constitutive closure

$$L_0(x) = 1 / \sqrt{(\mu_0^2 + a_\rho\, \rho_m(x) + b(x)|\nabla\chi(x)|^2)} \quad (5.3)$$

Here $\mu_0^2 = 2\lambda\Psi_0^2 = m_s^2$ is the vacuum scalar mass. The density-running term $a_\rho > 0$ has its sign fixed by the coupling, while the gradient term $b(x)|\nabla\chi|^2$ is model-dependent. The functional form (5.3) is therefore a constitutive closure rather than a result derived from the action.

5.3 Solver-Validated Results

L_0 (km)	M_eff/M_GR	ΔM (M⊙)	NICER?
10	0.268	−1.025	YES
50	0.862	−0.193	YES

L_0 (km)	M_eff/M_GR	ΔM (M⊙)	NICER?
100	0.948	−0.073	YES (marginal)
500	0.972	−0.039	NO (threshold)
1,000	0.988	−0.017	NO
∞ (GR)	1.000000	0.000	N/A

Table 3. Solver-corrected M_eff/M_GR for a 1.4 M⊙, R = 12 km neutron star (n = 1 polytrope). These results are solver-validated and should be treated as authoritative over earlier hand estimates.

The results of §5.1–§5.3 are action-derived or solver-validated. The following subsection introduces a constitutive closure for the density dependence of L_0. It is physically motivated and consistent with the constraints above, but it is not derived from the action.

5.4 Density-Running Closure (Model-Dependent)

NICER requires $L_0 < 500$ km at neutron-star scales, while Cassini requires $L_0 > 60{,}000$ km at solar-system densities. Simultaneously satisfying those constraints requires strong environmental running. A simple consistency fit is $L_0 \propto \rho^{(-\alpha)}$ with $\alpha \approx 0.1$, corresponding to a cross-scale ratio $L_0(\text{solar})/L_0(\text{NS}) \approx 120$. That ratio is an empirical consistency condition required to bridge the two observational scales, not a theoretical prediction of the framework. The exponent α is not derived from the action; it is a closure-dependent parameter whose value is set by the requirement that both bounds be simultaneously satisfied.

VI. Scalar Breathing Mode and Graviton Propagator

6.1 Breathing Mode Amplitude

For a circular binary in the CGG sector, the monopole source oscillates at twice the orbital frequency. The breathing-to-tensor ratio scales as

$$|h_B/h_+| \sim \eta^2 (GM/Rc^2) F(\alpha, \mu_s) \quad (6.1)$$

Here $\eta = v/c$. In the heavy-scalar regime, F is exponentially suppressed for $\alpha \mu_s > 1$, reflecting the evanescent character of the massive scalar. In the viable wedge this drives $|h_B/h_+|$ below current LIGO sensitivity. This suppression is regime-specific rather than unique to CGG.

The phrase "parametrically suppressed" therefore applies only in the heavy-scalar regime $m_s r_s \gtrsim 40$ with $\alpha \mu_s > 1$. It does not hold generically for light-scalar models.

6.2 QNM Shift Crossover Function

The QNM shift law (derivation in Appendix E):

$$|\Delta\omega/\omega| = \delta \cdot F(m_s r_s) \quad (6.2)$$

Heavy-scalar regime ($m_s r_s \gtrsim 40$ — the only viable CGG regime):

$$F(x) \approx C_1 x = 0.0495\, x, \quad x = m_s r_s \quad (6.3)$$

$C_1 = 0.0495$ (numeric, 200-point grid, plateau for $x > 60$). WKB cross-check ~0.055 (10% agreement; O(1) saddle-point systematic). At canonical anchor $\delta \cdot C_1 \cdot (m_s r_s) = 0.05 \times 0.0495 \times 71.8 \approx 0.178$.

The legacy quadratic form $F(x) \approx \frac{1}{2}x^2$ overestimates by ~720× at the canonical anchor and is incompatible with the viable parameter wedge. That correction is not cosmetic: without it, no viable wedge exists.

6.3 Tensor Propagator

At frequencies $\omega \gg m_s c^2/\hbar$, equivalently wavelengths $\lambda_{GW} \ll L_0$, the scalar field decouples from the tensor gravitational-wave sector. The tensor propagator then reduces to the standard massless spin-2 form.

> Scalar corrections to the tensor sector enter parametrically at order $(m_s/k)^2$, where k is the gravitational-wave momentum. For $k \gg m_s$ these corrections are negligible. (6.4)

A quantitative one-loop renormalization analysis of the non-minimal coupling sector has not been completed here, so radiative stability remains an open problem. The present claim is limited to classical decoupling in the heavy-scalar regime.

VII. Rotating Solutions and Vortex Kinematics

The rotating sector follows from phase structure already present in the complex scalar field. No new dynamical degrees of freedom are introduced, but quantitative rotating-sector predictions beyond phase kinematics remain less settled than the weak-field and compact-object sectors. Deriving the frame-dragging correction, vortex nucleation threshold, and critical spin from first principles requires solving the coupled Einstein-scalar system in the Hartle–Thorne slow-rotation expansion. That calculation is deferred to future work. The results presented in §7.2–7.4 are therefore indicative structural forms, not derived predictions.

7.1 Phase Decomposition and Conserved Current

Writing:

$$\psi = A\, e^{i\theta} \quad (7.1)$$

the theory carries a conserved U(1) current:

$$j^\mu \propto A^2\, \partial^\mu \theta \quad (7.2)$$

Rotation excites the phase sector already implicit in the covariant action. Quantized circulation and vortex-like interiors arise as allowed strong-field phase configurations of the original field — not as externally appended structure.

For azimuthal winding:

$$\oint \nabla\theta \cdot d\ell = 2\pi n, \qquad n \in \mathbb{Z} \quad (7.3)$$

Quantized circulation is a direct consequence of single-valuedness of ψ, not an external ansatz. The substrate velocity field:

$$v_\theta = (\hbar/m_s)\, \nabla\theta \quad (7.4)$$

is irrotational except at phase singularities (vortex cores), where $|\psi| \to 0$. Around a vortex core: $v_\theta(\rho_\perp) = n\hbar/(m_s\, \rho_\perp)$, the $1/\rho_\perp$ profile following directly from (7.3).

7.2 Frame-Dragging Correction (Indicative)

In a slowly rotating background, the leading correction to the Kerr frame-dragging frequency takes the structurally motivated form

$$\omega_{FD}(r) = \omega^{Kerr}_{FD}(r) \cdot [1 + (2\delta + \delta^2)\, H(r; m_s, r_s)] \quad (7.5)$$

Here $H(r_s) = 1$ and $H(r) \sim \exp(-m_s(r - r_s))$ for $r \gg r_s$, reflecting the finite range of the massive scalar. This form is indicative rather than derived: a complete Hartle–Thorne plus scalar calculation has not been carried out in this work. What is robust is the localization of the correction to a region of order $\xi = 1/m_s$ near the horizon.

7.3 Vortex Nucleation Threshold (Indicative)

Frame dragging enhances the coherence gradient near the horizon. The critical spin parameter for vortex nucleation has the parametric scaling

$$a_{crit}/M = (\varepsilon^* - \varepsilon_{b0})\, \xi^2 / (\eta\, m^2\, r_s\, |\nabla\chi|^2_{static}) \quad (7.6)$$

This expression should be read as an indicative scaling law rather than a fully derived threshold. A quantitative derivation would require the rotating interior solution of the coupled Einstein–scalar system, which is beyond the scope of the present paper.

7.4 Echo Mechanisms

Two distinct echo mechanisms arise from the same action in different limits. They are controlled by different parameters and have different status:

Property	Route 1 (exterior reflection)	Route 2 (interior cavity)
Controlling parameter	$\varepsilon = (r_c - r_s)/r_s$	$\xi/r_s = 1/(m_s r_s)$
Activation condition	All smooth-branch compact objects	Rotating objects: $a > a_{crit}$ only
Timing	$\Delta t = (2r_s/c)\ln(1/\varepsilon)$ [DERIVED]	$\Delta t \sim (r_s/c)\ln(r_s/\xi)$ [DERIVED]
Amplitude	$\exp(-2\int\sqrt{V_\ell - \omega_R^2}\,dr^*)$ [structural form settled; numeric TBD]	$\exp(-\pi\ell\,\xi/r_s)\cdot\delta$ [PROVISIONAL — see note]
$m_s r_s$ dependence	Grows as ε decreases (opposite)	Grows with m_s (opposite to R1)

Echo mechanisms. Timing predictions are derived from cavity geometry where the underlying cavity exists. Echo amplitudes are model-dependent because they depend on the interior coherence profile. Route 2 additionally depends on the indicative vortex-threshold condition of §7.3.

Both echo mechanisms are consequences of the same field equation in different limits:

Route 1 — Mode-sector tunneling (all compact objects with ε > 0)

$$R^{(1)}_{GW} \sim \exp\left(-2\int_{r_s}^{r_c}\sqrt{V_\ell - \omega_R^2}\,dr^*\right), \quad \Delta t^{(1)}_{echo} = (2r_s/c)\ln(1/\varepsilon) \quad (7.7)$$

Exists for all smooth-branch compact objects. Timing scales with $\ln(1/\varepsilon)$: grows as ε decreases.

Route 2 — Vortex Kelvin-mode coupling (a > a_crit only)

$$|h_{echo}/h_{QNM}|^{(2)} \sim \exp(-\pi\ell\,\xi/r_s)\,\delta, \quad \Delta t^{(2)}_{echo} \sim (r_s/c)\ln(r_s/\xi) \quad (7.8)$$

Key diagnostic: Route 1 and Route 2 have opposite dependences on $m_s r_s$. Simultaneous observation with different timing signatures would support the two-sector structure of CGG, though Route 2 remains contingent on the rotating-sector derivation.

VIII. Phenomenological Closures and Strong-Field Observables

The observables in this section depend on Tier II phenomenological closures (§1.1b) in addition to the action-derived core. Results are solver-validated but closure-dependent unless individually noted otherwise.

The framework is only meaningful if its sectors remain simultaneously viable under observation. The relevant falsification problem is not whether one individual sector can be made to fit data, but whether the same parameter wedge survives all of the following simultaneously:

- weak-field PPN recovery ($\gamma = 1$, $\beta = 1$),
- optical-clock bound on ε,
- compact-object QNM consistency,
- neutron-star coherence-length constraints,
- gravitational-wave polarization suppression, and
- Kerr-like exterior behavior for rotating objects at leading order.

That simultaneous wedge is the true viability region of CGG.

Observable	GR	CGG	Constraint	Derivation path
γ PPN	1	1.000 (derived §II.3)	$\lvert\gamma-1\rvert < 2.3\times10^{-5}$	Action → T μν → h ij/h00
β PPN	1	$1 + O(10^{-20})$	$\lvert\beta-1\rvert < 1.1\times10^{-4}$	Potential curvature
GW speed	c	c (exact)	$\lvert v/c-1\rvert < 10^{-15}$	Massless tensor propagator
Tidal k_2	0.05–0.15	$\sim10^{-12}$ (ε^2 suppressed)	$k_2 < 800$	Epsilon-bridge
QNM shift	GR	$\delta\cdot C_1\cdot(m\ s\ r\ s) \approx 0.18$	$\delta f/f < 3\%$	Corrected overlap integral
Echo R1 timing	None	$(2r_s/c)\ \ln(1/\varepsilon)$	$\Delta t > 10$ ms	Mode-sector gap
Echo R2 timing	None	$(r_s/c)\ \ln(r_s/\xi)$, a>a_crit	Rotation required	Vortex Kelvin mode
Breathing f_s	None	$\sim4f_{22}$ (heavy-scalar regime only)	Not detected	$\eta^2\cdot$compactness$\cdot$F
NS mass deficit	GR	$L_0 > 500$ km	NICER	Coherence weighting

Table 4. Master observable table. Entries descend from the central law (1.3) through the sector shown in the final column. 'Kerr-like' refers to leading-order exterior behavior only.

IX. Falsification Criteria

Observation	Implication for CGG	Instrument
Ringdown $\delta f/f < 10^{-12}$, Kerr-exact	Requires extreme fine-tuning of ε; effectively excluded	ET, LISA
Stacked echo $R > 0.01$ at $\Delta t \sim 10$–100 ms	Contradicts exponential suppression at $\varepsilon < 10^{-6}$	Advanced LIGO O5+
Love number $k_2 > 10$ in BNS merger	Contradicts ε^2 suppression	Advanced LIGO O5+
PPN $\lvert\gamma-1\rvert > 10^{-4}$ from solar ranging	Weak-field metric derivation breaks; theory excluded	Interplanetary ranging
NS masses within 2.8% of GR (NICER)	Constrains $L_0 > 500$ km at nuclear density	NICER
Cross-scale ratio L_0(solar)/L_0(NS) < 120	Density-running law (5.3) invalid	NICER + Cassini joint
Breathing mode at $f_s \sim 4f_{22}$, SNR > 5	Confirms the breathing mode prediction	Einstein Telescope
GW speed $\lvert v/c - 1\rvert > 10^{-15}$	Massless tensor propagator breaks	GW170817 + future events

Table 5. Falsification criteria. The theory is structurally falsifiable by instruments available within the next decade.

X. Discussion

10.1 How CGG Reduces to GR

Three independent limits each recover GR exactly:

- $\varepsilon \to 0$: mode-sector gap closes; all compact-object strong-field deviations vanish.
- $\delta \to 0$: smooth-branch deviation quenches; QNM and echo signatures disappear.
- $L_0 \to \infty$: coherence efficiency $\eta_{coh} \to 1$; universal Poisson equation fully restored.

The independence of these three limits explains how CGG satisfies PPN bounds (the third limit is sufficient) while maintaining distinct strong-field signatures (the other two limits have not been forced by current observations).

10.2 Relationship to Scalar-Tensor Theories

In the weak-field exterior with $\varepsilon \to 0$, CGG resembles a massive scalar-tensor theory in the limit $\omega_{BD} \to \infty$. Three features distinguish CGG structurally from standard scalar-tensor models:

- The mode-sector boundary at $r_c = r_s(1 + \varepsilon)$, absent in Brans-Dicke.
- The environment-dependent coherence length $L_0(\rho_m)$, breaking the universality of scalar-tensor coupling.
- The U(1) phase structure producing topological vortex defects with observable consequences only above a_{crit}.

10.3 Scope and Current Limitations

This paper establishes a unified effective field theory architecture, but not every constitutive element is equally settled. The most important open issues are the constitutive closure governing L_0, the absence of a full slow-rotation calculation, and the unresolved radiative stability of the non-minimal coupling sector.

XI. Limitations and Conclusions

11.1 Limitations

The present work establishes a unified effective field theory architecture. It does not claim completion of every constitutive closure required for quantitative predictions in the strong-field and rotating sectors.

11.2 Conclusions

This work introduces a unified coherence-field framework in which:

- macroscopic gravity arises from coherence energy gradients ($\nabla^2\varphi_c = 4\pi G\rho$ in the static limit);
- the Newtonian limit is recovered exactly through $\Phi_N = -c^2\delta C$;
- compact-object interiors follow smooth-branch scalar solutions derived from the same action; and
- observable deviations appear through coherence-length physics (L_0, ε) constrained by NICER, Cassini, and optical clocks.

We have presented Coherence Gradient Gravity as a unified scalar effective field theory formulated in a generally covariant framework, with weak-field gravity, compact-object structure, and coherence-length physics descending from a common action.

The essential point is not merely that the theory contains these sectors, but that they are mathematically linked. The scalar amplitude $C = |\psi|^2$ produces the weak-field gravitational potential; the same nonlinear envelope governs compact-object structure; the same effective mass scale controls coherence-length running; and the same phase degree of freedom generates rotating-current and vortex sectors. No later extension requires new field content external to the original action.

In the weak-field static limit the theory recovers the standard Poisson equation exactly through the identification $\Phi_N = -c^2\,\delta C$, yielding the correct Newtonian acceleration law and post-Newtonian metric structure. In the strong-field regime, smooth-branch compact objects remain viable only in the heavy-scalar wedge ($m_s r_s = 70$–100, $\delta = 0.02$–0.05) required by the current empirical bound on ε. The rotating sector preserves Kerr-like exterior behavior at leading order while admitting interior phase structures that remain observationally testable.

The remaining uncertainties are constitutive and phenomenological closure problems inside an already unified framework. They should be treated as such: not as proof of internal inconsistency, but as explicit model-dependent elements awaiting derivation, numerical closure, or observational exclusion.

1. Central law. All sectors derive from $\nabla^2(\delta C) = -\kappa\rho$ under different conditions on η_{coh}. This equation is the spine of the paper.
2. Metric derivation. $h_{00} = +2\delta C$ and $h_{ij} = +2\delta C\,\delta_{ij}$ are derived from the scalar stress-energy in the static weak-field limit. $\gamma_{PPN} = 1$ holds at leading SVEA order.

3. Newtonian closure. $\Phi_N \equiv -c^2\delta C$ gives $\nabla^2\Phi_N = 4\pi G\rho$ and $a = -GM/r^2\,\hat{r}$ exactly, from the field equation alone.
4. Parameter reduction. After $g_m\Psi_0^2 = 4\pi G$ fixes the coupling, the residual freedom is m_s, δ, and the L_0 closure function. The carrier frequency $\omega_c = m_s c^2/\hbar$ is exact under SVEA normalization — not a dimensional estimate.
5. Vortex kinematics from the action. The U(1) phase structure generates $v_\theta = (\hbar/m_s)\nabla\theta$. Vortices are topological defects of θ. Frame dragging and a_{crit} retain only indicative status pending a full slow-rotation calculation; Route 2 amplitudes remain provisional.
6. Two echo routes, one field equation. Route 1 timing $\propto \ln(1/\varepsilon)$; Route 2 timing $\propto \ln(r_s/\xi)$. Opposite $m_s r_s$ dependences; distinguishable in principle.
7. QNM shift corrected. $F(x) = C_1 x$ with $C_1 = 0.0495$ replaces the legacy quadratic form (which overestimates by ~720× at the canonical anchor). This correction is what makes the viable wedge exist.
8. Simultaneous viability wedge ($m_s r_s = 70$–100, $\delta = 0.02$–0.05, $L_0 > 500$ km). The anchor-point ε bound and the neutron-star Poisson results are solver-validated; the solar bound is empirical; the density-running dynamic range remains closure-dependent. The wedge is therefore consistent but not uniformly derived from first principles.

Appendix A: Weak-Field Recovery in Four Lines

This appendix makes the GR limit explicit in compact form.

Starting from the central law:

$$\nabla^2(\delta C) = -\kappa\rho, \quad \kappa = 4\pi G/c^2 \quad \text{(A.1)}$$

Define:

$$\boldsymbol{\Phi_N \equiv -c^2\,\delta C} \quad \text{(A.2)}$$

Then:

$$\nabla^2\Phi_N = -c^2\,\nabla^2(\delta C) = -c^2(-\kappa\rho) = 4\pi G\rho \quad \text{(A.3)}$$

The Newtonian Poisson equation is recovered exactly. The geodesic equation gives:

$$a = -\nabla\Phi_N \quad \text{(A.4)}$$

For $\Phi_N = -GM/r$:

$$a = -(GM/r^2)\,\hat{r} \quad \text{(A.5)}$$

Newtonian gravity follows from the field equation alone. No additional identification is required beyond (A.2).

Appendix B: Liouville-Transformed Radial Operator

Starting from phase perturbation in spherical symmetry:

$$d/dr[r^2\Psi^2\theta'(r)] + [\omega^2 r^2\Psi^2 - \ell(\ell+1)\Psi^2]\theta(r) = 0 \quad \text{(B.1)}$$

Define $u(r) = r\Psi(r)\theta(r)$. Substitution removes the first-derivative term:

$$u''(r) + [\omega^2 - V_\ell(r)]\,u(r) = 0 \quad \text{(B.2)}$$

Envelope contribution to V_ℓ: $V_env(r) = -1/r^2 + \frac{1}{2}\chi''(r)$, where $\chi = \ln C = 2\ln\Psi$.

Appendix C: Radial Solver Validation

- GR recovery: At $L_0 = 316{,}000$ km, $M_eff/M_GR = 1.000000$ (six decimal places). Exterior $r\cdot\delta C$ constant to machine precision.
- Mass bound: $M_eff/M_GR \leq 1.000$ for all $L_0 \in [0.3, 316{,}000]$ km.
- Grid convergence: 1500→3000 points changes M_eff/M_GR by $< 10^{-6}$.

Solver completes in < 1 ms (single-threaded, 3000 points).

Appendix D: Coupled Einstein-Scalar ODE System

State vector (M, Φ, Ψ, Π); full autonomous ODE system:

$$dM/dr = 4\pi r^2[\tfrac{1}{2}f(\Psi')^2 + V(\Psi)] \quad \text{(D.1)}$$

$$d\Phi/dr = [GM/r^2c^2 + (4\pi Gr/c^4)(\tfrac{1}{2}f(\Psi')^2 - V(\Psi))]/f \quad \text{(D.2)}$$

$$d\Psi/dr = \Pi \quad \text{(D.3)}$$

$$d\Pi/dr = -(2/r + \Phi' + f'/(2f))\Pi + V'(\Psi)/f \quad \text{(D.4)}$$

Near-horizon BC (4.2). Asymptotic BC: $\delta\Psi \sim r^{-1} \exp(-m_s r)$.

Appendix E: Heavy-Scalar Asymptotics of the QNM Shift Integral

E.1 Potential Perturbation

For the smooth branch with Yukawa falloff $\delta\Psi(r) = \delta\Psi_0(r_s/r)\exp(-m_s(r-r_s))$:

$$\Delta V(r) \sim \delta\, m_s^2 f(r) \exp(-2m_s(r-r_s)) \quad \text{(E.1)}$$

E.2 First-Order QNM Shift Functional

$$\Delta\omega = \int \Delta V(r^*)\, \psi^2_{QNM}(r^*)\, dr^* / [2\omega \int \psi^2_{QNM}(r^*)\, dr^*] \quad \text{(E.2)}$$

E.3 Overlap Integral (Heavy-Scalar)

Integration dominated by the Yukawa shell near r_s:

$$\int \Delta V\, dr^* \sim \delta\, m_s^2 \cdot (1/m_s) \cdot |\psi_{QNM}(r_s)|^2 \cdot r_s = \delta\, m_s\, |\psi_{QNM}(r_s)|^2\, r_s \quad \text{(E.3)}$$

Hence:

$$|\Delta\omega/\omega| = \delta \cdot C_1 \cdot (m_s r_s), \qquad C_1 = m_s r_s |\psi_{QNM}(r_s)|^2 / (2\omega_R \int \psi^2_{QNM}\, dr^*) \quad \text{(E.4)}$$

E.4 Legacy vs. Corrected

$m_s r_s$	$F_{corr} = C_1 x$	$F_{legacy} = 0.5x^2$	Overestimate	Note
10	0.60	50	83×	borderline
62	3.09	1922	622×	ε-bridge threshold
71.8	3.55	2575	723×	canonical anchor
100	4.95	5000	1010×	deep heavy-scalar

Table E1. The legacy quadratic form overestimates by ~720× at the canonical anchor. The corrected linear scaling makes the viable wedge possible.

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